



Internship in

Artificial Intelligence

Codec Technologies

Aditya Prakash
25SCS1003005350
B.Tech CSE, 3rd Semester

2026-27

Candidate & Program Overview

Candidate Details

Student Name: ADITYA PRAKASH

Roll Number: 25SCS1003005350

Degree: B.Tech (Computer Science & Engineering)

Academic Batch: 2026-2027

University: IILM University, Greater Noida, U.P.

Internship Profile

Program Name: Artificial Intelligence Internship

Specialization Domain: Artificial Intelligence

Duration: 4 Weeks (August 2026 – September 2026)

Organization Profile: Codec Technologies

Codec Technologies is an IT consultancy, technology services, and educational training organization. The organization provides IT solutions including web and application development, cloud hosting, software deployment, infrastructure management, and digital transformation services.

Codec Technologies also provides corporate training and internship programs in various technical domains such as Artificial Intelligence, Data Science, Python Development, Cyber Security, Web Development, and Data Analytics. Its internship programs are designed to provide learners with structured training and practical exposure to industry-related technologies.



Problem Statement

The internship aimed to address the need for a structured introduction to these fundamental AI concepts and their practical applications. The challenge was to understand the complete workflow of an AI-based solution, beginning with data preparation and feature extraction and progressing toward model development and evaluation.

As a practical application of the concepts learned, the project component focused on developing a movie recommendation system capable of identifying movies similar to a selected movie based on its content and metadata.

Project Objectives

- To understand the fundamental concepts and applications of Artificial Intelligence.
- To develop basic programming and data-handling skills using Python.
- To learn techniques for data preprocessing, visualization, and preparation for AI applications.
- To understand the fundamentals of supervised learning and machine learning models.
- To learn the basics of model evaluation and performance measurement.
- To gain an introductory understanding of neural networks and deep learning.
- To explore Natural Language Processing and basic text-processing techniques.
- To understand the basic principles and applications of Reinforcement Learning and Q-learning.
- To apply the concepts learned during the internship through a practical AI project.

Technologies

Python

NumPy

Pandas

Scikit-learn

Jupyter Notebook / Google Colab

Project & Demonstrated Skills

The architecture of the Content-Based Movie Recommendation System begins with a movie dataset containing information such as movie titles, genres, descriptions, keywords, and other metadata. The data is first loaded and preprocessed to handle missing or irrelevant information. Relevant attributes are then selected and transformed into a suitable feature representation.

The extracted features are converted into numerical vectors, which allow the system to mathematically compare movies. Similarity between movies is calculated using cosine similarity, after which the movies are ranked according to their similarity scores. Finally, the highest-ranked movies are presented as recommendations.



Project & Demonstrated Skills

```
interactive_recommend()
```

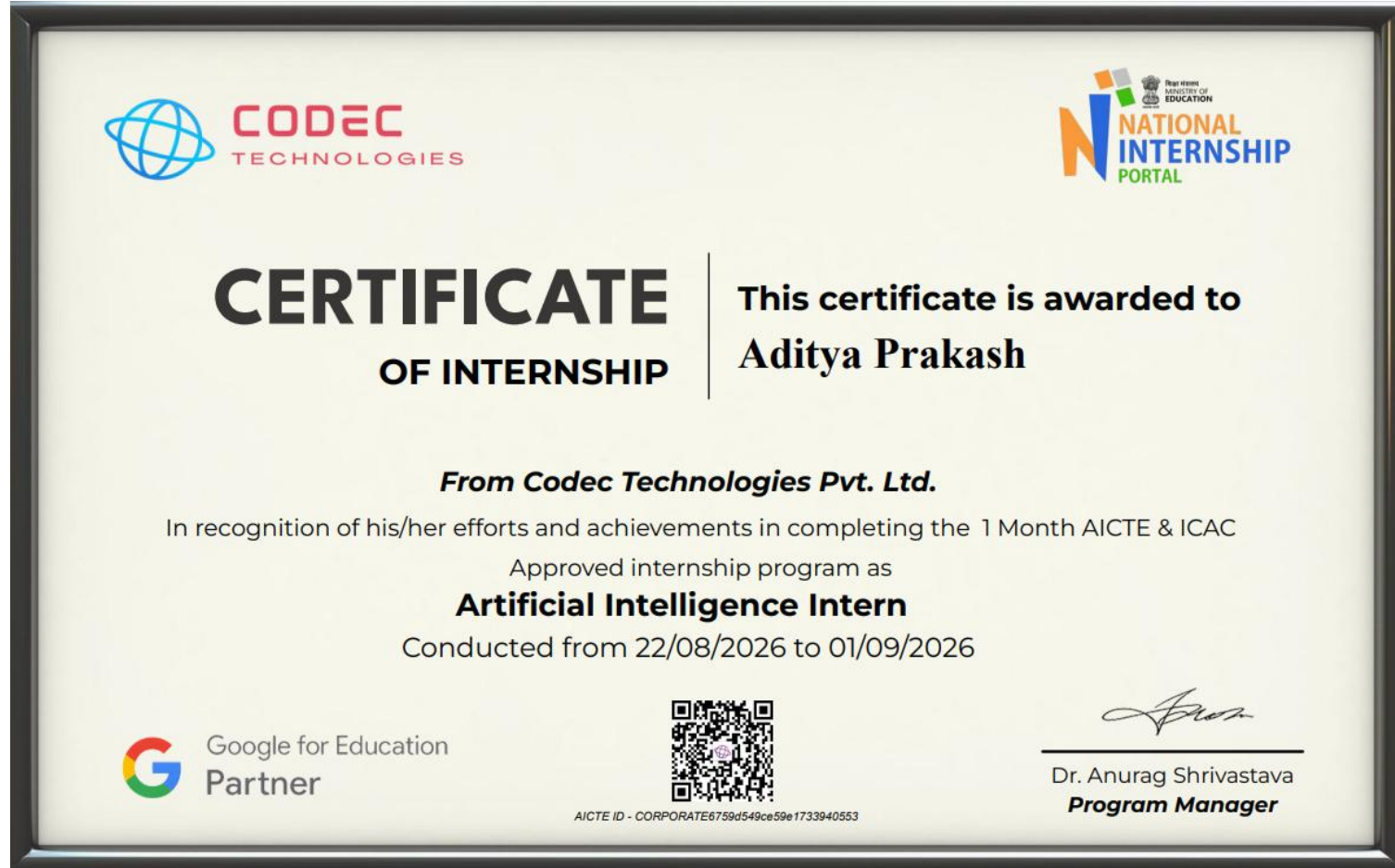
```
[10] ✓ 7.6s
```

```
...
```

```
Top Recommendations for: 'The Dark Knight'
```

```
-----  
1. The Dark Knight Rises          | Similarity: 0.472  
2. Batman Begins                 | Similarity: 0.418  
3. Batman Returns                | Similarity: 0.313  
4. Batman                       | Similarity: 0.260  
5. Batman: The Dark Knight Returns, Part 2 | Similarity: 0.256  
6. Batman & Robin                | Similarity: 0.225  
7. Batman Forever                | Similarity: 0.222  
8. Defendor                     | Similarity: 0.187  
9. Batman v Superman: Dawn of Justice | Similarity: 0.185  
10. The Prestige                 | Similarity: 0.180
```

Certification



Thank You!